LLM Evaluation Report

Loan Default Prediction | LendingClub 2012-2014 | Capstone Project

1. Experimental Setup

This report summarises three experiments evaluating Large Language Models (LLMs) as zero-shot loan default classifiers on LendingClub data. Each experiment tests a 100-loan sample drawn from the held-out test set (2012-2014 originations). Loans are binary-classified as **Fully Paid (1)** or **Charged Off (0)**. Every LLM receives the same structured features available to the ML baseline (XGBoost with Optuna-tuned threshold), and optionally a free-text borrower description (*desc*) written at application time.

Two conditions are tested per model: **no_desc** (structured features only) and **with_desc** (structured features + borrower description). The XGBoost baseline uses structured features only and serves as the ML benchmark.

2. Experiment 04a — Model Comparison

Three LLMs were evaluated against XGBoost on the same 100-loan sample (84 Fully Paid, 16 Charged Off):

Model	Condition	Accuracy	CO Precision	CO Recall	CO F1
XGBoost	structured	71%	29.0%	56.3%	0.383
Gemini 2.5 Flash	no_desc	61%	24.4%	68.8%	0.361
Gemini 2.5 Flash	with_desc	58%	19.0%	50.0%	0.276
Gemini 2.5 Pro	no_desc	58%	24.0%	75.0%	0.364
Gemini 2.5 Pro	with_desc	57%	21.3%	62.5%	0.317
GPT-5	no_desc	73%	26.1%	37.5%	0.308
GPT-5	with_desc	80%	38.9%	43.8%	0.412

Table 1. 04a Model Comparison — accuracy and Charged Off metrics.

Key findings:

- **GPT-5 with_desc achieves 80% accuracy**, surpassing XGBoost (71%) by 9 percentage points and all other LLM configurations by at least 7 points.
- **Description effect is model-dependent.** Descriptions *help* GPT-5 (+7 pp accuracy, CO F1 0.31 → 0.41) but *hurt* both Gemini models, which over-react to negative language and over-predict defaults.
- **Calibration gap.** Gemini models predict 42-50 defaults (vs 16 actual); GPT-5 predicts 18 — nearly perfectly calibrated.

3. Experiment 04b — Consistency

GPT-5 was run **three times per condition** on the same 100-loan sample to measure output stability. Because LLMs are stochastic, we need to verify that results are reproducible rather than artifacts of a lucky run.

Condition	Run 1	Run 2	Run 3	Mean	Std
no_desc	79%	77%	79%	78.3%	1.15%
with_desc	78%	80%	80%	79.3%	1.15%

Table 2. 04b Accuracy across 3 repeated runs per condition.

Condition	Run 1	Run 2	Run 3	Range
no_desc	0.400	0.378	0.400	0.378 - 0.400
with_desc	0.389	0.375	0.412	0.375 - 0.412

Table 3. 04b Charged Off F1-score across repeated runs.

Prediction Stability

Condition	Stable (3/3 agree)	Unstable	Stability %
no_desc	92	8	92%
with_desc	93	7	93%

Table 4. 04b Prediction agreement across 3 runs (100 samples).

Key findings:

- **Highly reproducible.** Accuracy varies by only ±1% across runs in both conditions (std = 1.15%).
- **92-93% prediction stability.** GPT-5 gives the identical answer on 92-93 out of 100 loans regardless of the run. The 7-8 unstable samples represent genuinely ambiguous borderline cases.
- **Charged Off recall is rock-solid** at 43.8% (7/16) in 5 of 6 runs. The model consistently identifies the same set of defaulting loans.
- **with_desc edges out no_desc** on average (79.3% vs 78.3% accuracy; best CO F1 0.412 vs 0.400), confirming the description benefit is not random noise.

4. Experiment 04c — Robustness

GPT-5 was evaluated on a **completely new, non-overlapping 100-loan sample** drawn from the same test set (77 Fully Paid, 23 Charged Off — a harder class distribution than the original 84/16 split).

Model	Condition	Accuracy	CO Precision	CO Recall	CO F1
XGBoost	structured	70%	39.4%	56.5%	0.464
GPT-5	no_desc	72%	35.3%	26.1%	0.300
GPT-5	with_desc	76%	46.2%	26.1%	0.333

Table 5. 04c Robustness — new 100-loan sample results.

Original vs New Sample

Model	Condition	Orig Acc	New Acc	Δ
XGBoost	structured	71%	70%	-1 pp
GPT-5	no_desc	73%	72%	-1 pp
GPT-5	with_desc	80%	76%	-4 pp

Table 6. Accuracy comparison: original sample (04a) vs new sample (04c).

Key findings:

- **Model ranking is preserved.** GPT-5 with_desc (76%) > GPT-5 no_desc (72%) > XGBoost (70%) holds on an entirely independent sample.
- **Description effect generalises.** with_desc gains +4 pp accuracy over no_desc, and CO precision jumps from 35% to 46% — the description helps GPT-5 make more confident, correct default calls on unseen data.
- **Magnitude is sample-dependent.** The new batch has 23 defaults (vs 16), making it harder. The 4 pp drop for with_desc (80% → 76%) is expected, while XGBoost remains stable (71% → 70%) as a trained model.

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5. Conclusions

Across all three experiments, GPT-5 with access to borrower descriptions consistently outperforms the Optuna-tuned XGBoost baseline in overall accuracy. The key findings are:

- **GPT-5 is the only LLM that benefits from unstructured text.** Both Gemini models over-react to negative language in descriptions, worsening their predictions. GPT-5 interprets the same text as evidence of financial responsibility.
- **Results are reproducible** (±1% accuracy, 92-93% prediction stability) and **robust** to sample variation (ranking preserved on independent data).
- **Trade-off exists in default detection.** XGBoost catches more defaults (higher CO recall) but at the cost of more false alarms. GPT-5 is more precise when it flags a default, making fewer but more reliable calls.
- **LLMs offer interpretability.** Every GPT-5 prediction includes a natural-language reasoning trace, providing transparency that black-box ML models lack.

Report generated from experiments 04a (Model Comparison), 04b (Consistency), and 04c (Robustness). All code and results are available in the project repository.